

Secrecy Capacity of 1-2-1 Atomic Networks with Balanced Atoms

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Abstract

This work studies the information-theoretic secrecy capacity of 1-2-1 networks, a noiseless model introduced to capture the high-directionality of millimeter wave communication. The focus is on 1-2-1 networks with a single source, a single destination, and three layers of intermediate nodes. The nodes in the second layer represent a bottleneck for the communication and are referred to as *atoms*. It is assumed that these atoms are *balanced*, that is, they have a similar workload which is measured in terms of paths passing through them. The main contribution of this work consists of characterizing the maximum amount of information that can flow from the source to the destination over such networks and kept secret from the eyes of an eavesdropper, who can intercept the transmissions over a number of network links of their choice. This is accomplished by deriving a novel outer bound on the secrecy capacity and by designing a secure transmission scheme that suitably distributes traffic across the different paths connecting the source to the destination.

I. INTRODUCTION

Millimeter Wave (mmWave) communication is a key enabler for achieving high data rates and low-latency, which are essential for powering data-intensive applications, such as augmented reality, high-definition video streaming, and massive IoT deployments. A defining characteristic of mmWave communication is high-directionality: mmWave nodes are equipped with high-gain directional antenna arrays that can form narrow beams to direct the transmission and reception of signals in specific directions. With the two-fold objective of modeling this high-directionality in a simple yet informative way and studying the information-theoretic capacity of mmWave networks, the authors of [1] introduced the *1-2-1 network model*. The key characteristic of this model is that a link from node i to node j is activated if and only if node i points its transmit beam(s) towards node j and node j points its receive beam(s) towards node i . This beam scheduling can therefore be leveraged to activate different sets of links at different time instances. As pointed out in [2] and [3], this also holds promises for establishing a communication between two legitimate nodes that remains *secure* from the eyes of an eavesdropper who is physically present over a set of network links and hence, can intercept transmissions over them. In particular, this beam scheduling allows one to use a portion of the network links to transmit random packets (i.e., keys) meanwhile another portion can be used to

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transmit the data packets encoded with the keys. This represents a way to suitably leverage *multipath* to establish a secret communication between two legitimate parties. Multipath has recently been employed as a tool for enhancing security in noiseless networks within the framework of secure network coding, see for instance [4–9].

In this paper, we study the information-theoretic secrecy capacity of 1-2-1 networks, where a source wishes to communicate to a destination in the presence of an eavesdropper with unbounded computational capabilities, but limited network presence, i.e., the adversary can eavesdrop at most K network links of their choice (and hence, fully intercept all the transmissions over these links) on which they are physically present. The focus is on a class of 1-2-1 networks with a layered structure where the communication links are lossless and of unitary capacity. These are referred to as 1-2-1 *atomic* networks [10]. An example of such a network is given in Figure 1, where the nodes v_1 , v_2 , and v_3 are referred to as *atoms*. In this network, at any point in time, the source and the destination can direct their transmit and receive beams towards at most $M \geq 1$ nodes, whereas all the other nodes can (simultaneously) receive from and transmit to at most another node. In other words, the source and the destination have more advanced beamforming capabilities than the other nodes. The characterization of the information-theoretic secrecy capacity of 1-2-1 atomic networks has already been our focus in [10]. In particular, in [10] we derived an upper bound and a lower bound on the secrecy capacity of 1-2-1 *atomic* networks and provided conditions under which they match. However, when such conditions are not satisfied, then the secrecy capacity is not known. Our main contribution in this work consists of characterizing the secrecy capacity of 1-2-1 *atomic* networks in a regime that was left open in [10]. Broadly speaking, this consists of having atoms with a similar “workload”¹ and hence, we refer to these networks as *1-2-1 atomic networks with balanced atoms*. We derive a new outer bound on the secrecy capacity and we design a novel secure transmission scheme that achieves a secrecy rate that matches the derived upper bound, hence characterizing the secrecy capacity of 1-2-1 atomic networks with balanced atoms. One key feature that differentiates the designed scheme from the one that we proposed in [10] is that we allow the same link to be used multiple times for transmission. Our work falls within the framework of secure network coding on which the literature is extensive, with notable contributions including [11–17]. However, these works typically assume network models in which nodes can broadcast to all their connected peers simultaneously. In contrast, the present work operates under a *directivity* constraint - specifically, the 1-2-1 constraint - wherein each node is restricted to communicating with only a limited subset of its neighbors.

Paper organization. In Section II, we review the 1-2-1 atomic network that we introduced in [10] and we outline some known secrecy capacity results for 1-2-1 networks. In Section III, we present our main result, i.e., the secrecy capacity of 1-2-1 atomic networks with balanced atoms. In the same section, we also provide the outline of the proof of the outer bound and the design of the secure transmission scheme. Some of the proofs can be found in the Appendix.

¹As it will become clear in the paper, the workload of an atom is measured by the number of edge disjoint paths from the source to the destination that pass through that atom.

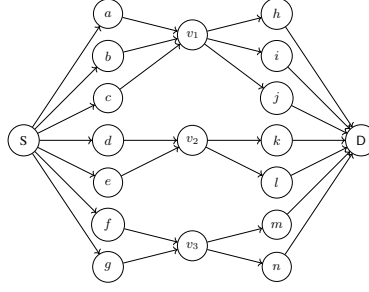


Figure 1: Example of a 1-2-1 atomic network with three atoms v_1 , v_2 , and v_3 .

II. SYSTEM MODEL AND BACKGROUND

Notation. Sets are denoted by calligraphic letters (e.g., $\mathcal{A}$) and vectors are denoted by boldface letters (e.g., $\mathbf{a}$). We use $\mathbf{a} \oslash \mathbf{b}$ to denote the Hadamard (element-wise) division of $\mathbf{a}$ by $\mathbf{b}$, i.e., $\mathbf{a} \oslash \mathbf{b} = \left[\frac{a_1}{b_1} \quad \dots \quad \frac{a_n}{b_n} \right]$. For any $m \in \mathbb{N}$, we define $[m] := \{1, 2, \dots, m\}$; $[a : b]$ is the set of integers from a to $b \geq a$. For a set $\mathcal{X}$, $|\mathcal{X}|$ denotes its cardinality; $\emptyset$ is the empty set. For a real number $r \in \mathbb{R}$, we denote its floor by $\lfloor r \rfloor$. We use $T_{\mathcal{A}}$ to denote $\{T_j : j \in \mathcal{A}\}$. $\square$

We study a 1-2-1 network in which a source node S seeks to establish secure communication with a destination node D . The communication network is represented by a directed acyclic graph $G = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of vertices such that $(S, D) \in \mathcal{V}$ and $\mathcal{E}$ is the set of edges. Each edge is assumed to be noiseless and to have unitary capacity. Communication across the network adheres to the 1-2-1 activation constraints [1]. Specifically, every intermediate node $v \in \mathcal{V} \setminus \{S, D\}$ operates in full-duplex mode and, at any given time instance, it may only receive from a single incoming edge and transmit over a single outgoing edge. For example, with reference to Figure 1, at any given time, node v_2 can receive from at most one among nodes d and e , and it can transmit to at most one among nodes k and l . In contrast, the source node S (respectively, the destination node D) is allowed to transmit to (respectively, receive from) up to $M \geq 1$ nodes concurrently. We restrict our attention to a structured subclass of 1-2-1 networks, referred to as *1-2-1 atomic networks with balanced atoms*, which we formally define next.

Definition 1 (1-2-1 Atomic Network [10]). *Let $G = (\mathcal{V}, \mathcal{E})$ be a directed acyclic graph with two distinct nodes $S, D \in \mathcal{V}$. We say that G is a 1-2-1 atomic network if it can be divided into subgraphs $\{G_i = (\mathcal{V}_i, \mathcal{E}_i)\}_{i=1}^{H_v}$, with $\bigcup_i \mathcal{V}_i = \mathcal{V}$ and $\bigcup_i \mathcal{E}_i = \mathcal{E}$ such that the following conditions hold for each $i \in [H_v]$:*

- 1) *The maximum number of **vertex-disjoint** (S, D) -paths in G_i is equal to 1.*
- 2) *The maximum number of **edge-disjoint** (S, D) -paths in G_i is equal to $h_i \in \mathbb{N}$, and all such paths pass through a common intermediate node $v_i \in \mathcal{V}_i \setminus \{S, D\}$, which is referred to as an atom of G .*
- 3) *$\mathcal{V}_i \cap \mathcal{V}_j = \{S, D\}$ for all $i \neq j \in [H_v]$.*

We denote such a network by a vector $\mathbf{h} = [h_1, \dots, h_{H_v}]$, where H_v denotes the maximum number of vertex-disjoint paths in the network. Moreover, we let $H_e = \sum_{i=1}^{H_v} h_i$, that is, H_e is the maximum number of edge-disjoint paths in the network. Without loss of generality, we assume that $h_1 \geq h_2 \geq \dots \geq h_{H_v}$.

Definition 2 (m -balanced 1-2-1 atomic network). Let G be a 1-2-1 atomic network with $\mathbf{h} = [h_1, \dots, h_{H_v}]$ and $H_e = \sum_{i=1}^{H_v} h_i$. For $m \in \mathbb{N}$, we say that G is m -balanced if $h_1 \leq \frac{H_e}{m}$.

Example 1. Consider the 1-2-1 atomic network in Figure 1. For this network, we have that $H_v = 3$ and $\mathbf{h} = [3, 2, 2]$; hence, $H_e = 7$. This network is 2-balanced since $h_1 = 3 < 7/2$.

The communication from S to D over a 1-2-1 atomic network occurs in the presence of an *external passive adversary* who can eavesdrop any K edges of their choice (unknown to all the other parties in the network). If the adversary eavesdrops edges in $\mathcal{Z} \subseteq \mathcal{E}$ with $|\mathcal{Z}| = K$, we require that the communication remains secure from the adversary in the following sense,

$$I(W; T_{\mathcal{Z}}^{[n]}) \leq \varepsilon, \quad \forall \mathcal{Z} \subseteq \mathcal{E}, \quad |\mathcal{Z}| = K, \quad (1)$$

where W is the message with entropy rate R that S wishes to transmit to D and $T_{\mathcal{Z}}^{[n]} = \{T_e^{[n]} : e \in \mathcal{Z}\}$ denotes the packets transmitted over $e \in \mathcal{Z}$ in n network uses.

In this paper, we wish to characterize the secrecy capacity C_s of 1-2-1 atomic networks. The secrecy capacity C_s is defined as the maximum rate at which S can communicate to D with zero error, while satisfying the 1-2-1 constraints of such networks and the security constraint in (1). In particular, the following results on the secrecy capacity C_s for 1-2-1 atomic networks are available in the literature:

- $M = 1$: From [2, Theorem 1, Theorem 2], we have that

$$C_s = \frac{H_e - K}{H_e}. \quad (2)$$

- $H_v \leq M$: From [10, Theorem 3], we have that

$$C_s = \sum_{i=1}^{H_v} \left(1 - \frac{K_i}{h_i}\right), \quad (3a)$$

where

$$K_i = \min \left\{ h_i, K - \sum_{j=i+1}^{H_v} K_j \right\}, \quad i \in [H_v]. \quad (3b)$$

From the above results, it therefore follows that the secrecy capacity C_s for 1-2-1 atomic networks is not known when $1 < M < H_v$. For this regime, in fact, only upper and lower bounds on C_s are available and these are not tight [2, 10]. Our main contribution is this work consists of characterizing the secrecy capacity C_s for M -balanced 1-2-1 atomic networks when $1 < M < H_v$. In other words, we characterize C_s for 1-2-1 atomic networks when $1 < M < H_v$ and $h_1 \leq H_e/M$. We conclude this section by providing two examples that show how our results improve over existing ones.

Example 2. Consider a 1-2-1 atomic network with $\mathbf{h} = [3, 2, 2]$ (as the one in Figure 1), $M = 2$, and $K < 7$ eavesdropped edges. By [2, Theorem 2], we get $C_s \geq 2 \left(1 - \frac{K}{3}\right)$ and by [10, Theorem 1] we have that $C_s \geq \max \left\{ \frac{6-K}{3}, \frac{7-K}{4} \right\}$. In this work, we will prove that $C_s = 2 \left(1 - \frac{K}{7}\right)$.

Example 3. Consider a 1-2-1 atomic network with $\mathbf{h} = [4, 3, 2]$, $M = 2$, and $K = 5$. By [2, Theorem 3], we have that $C_s \leq \frac{8}{9}$ and by [2, Theorem 2], we have that $C_s \geq 0$. In this work, we will indeed prove that $C_s = \frac{8}{9}$.

III. MAIN RESULT

In this section, we derive the secrecy capacity C_s of M -balanced 1-2-1 atomic networks when $1 < M < H_v$. We do this by deriving matching upper and lower bounds on C_s . The following theorem provides the main result of our work.

Theorem 1. *For an M -balanced 1-2-1 atomic network with K eavesdropped edges with $1 < M < H_v$ and with (without loss of generality) $h_1 \geq h_2 \geq \dots \geq h_{H_v}$, it holds that*

$$C_s = M \left(1 - \frac{K}{H_e} \right). \quad (4)$$

Proof: The proof consists of deriving an upper bound on C_s and a lower bound on C_s and showing that these two bounds match. The upper bound is derived in Section III-A and the lower bound is derived in Section III-B. ■

A. Outer Bound on C_s

We here show that the right-hand side of (4) is an upper bound on the secrecy capacity of an M -balanced 1-2-1 atomic network with $1 < M < H_v$. Toward this end, we start by deriving an upper bound on C_s for an *arbitrary* 1-2-1 atomic network (i.e., not necessarily M -balanced and not necessarily with $1 < M < H_v$). This outer bound is tighter than the upper bound that we derived in [10]. Our new upper bound is given by the following theorem.

Theorem 2. *For a 1-2-1 atomic network with K eavesdropped edges, let $\widehat{M} := \min(M, H_v)$ and assume (without loss of generality) that $h_1 \geq h_2 \geq \dots \geq h_{H_v}$. Then, it holds that*

$$C_s \leq \max_{\alpha \in \mathcal{A}_{\widehat{M}}} \sum_{i=1}^{H_v} \alpha_{\pi(i)} \left(1 - \frac{K_{\pi(i)}^*}{h_{\pi(i)}} \right), \quad (5)$$

where $\mathcal{A}_{\widehat{M}} = \left\{ \alpha \in [0, 1]^{H_v} : \sum_{i=1}^{H_v} \alpha_i = \widehat{M} \right\}$, $\pi(i)$ is the i -th maximum element in the vector $\alpha \odot \mathbf{h}$, and $K_{\pi(i)}^* = \min \left\{ K - \sum_{j=1}^{i-1} K_{\pi(j)}^*, h_{\pi(i)} \right\}$.

Proof: The proof is in Appendix A. ■

Remark 1. We highlight that $\sum_{i=1}^{H_v} K_{\pi(i)}^* = K$. We also note that $K_{\pi(i)}^*, i \in [H_v]$, in Theorem 2 can be rewritten as follows: There exists a $t \in [H_v]$, such that

$$K_{\pi(i)}^* = \begin{cases} h_{\pi(i)} & \text{if } i < t, \\ K - \sum_{j=1}^{i-1} h_{\pi(j)} & \text{if } i = t, \\ 0 & \text{if } i > t. \end{cases} \quad (6)$$

The upper bound on the secrecy capacity C_s in Theorem 2 still involves solving an optimization problem. In what follows, we solve it for the two cases $\widehat{M} = H_v$ (i.e., $H_v \leq M$), and $\widehat{M} = M$ when $h_1 \leq \frac{H_e}{M}$ (i.e., the M -balanced case with $1 < M < H_v$). In particular, in the next corollary, we consider the case $\widehat{M} = H_v$ (i.e., $H_v \leq M$), which indeed leads to an upper bound on C_s that matches the secrecy capacity that we proved in [10, Theorem 3].

Corollary 1. Assume that the conditions in Theorem 2 hold. Furthermore, assume $\widehat{M} = H_v$. Then, it holds that

$$C_s \leq \sum_{i=1}^{H_v} \left(1 - \frac{K_i^*}{h_i}\right), \quad (7)$$

where $K_i^* = \min \left\{ K - \sum_{j=i+1}^{H_v} K_j^*, h_i \right\}, i \in [H_v]$.

Proof: If $\widehat{M} = H_v$, the set $\mathcal{A}_{\widehat{M}}$ in Theorem 2 becomes $\mathcal{A}_{H_v} = \left\{ \alpha \in [0, 1]^{H_v} : \sum_{i=1}^{H_v} \alpha_i = H_v \right\}$ and hence, α such that $\alpha_i = 1, \forall i \in [H_v]$ is the only feasible point in $\mathcal{A}_{H_v}$, i.e., the solution for (5). Moreover, with the assumption of the h_i 's in descending order, we have that $\pi(i) = H_v - i + 1, \forall i \in [H_v]$. Substituting $\alpha_i = 1, \forall i \in [H_v]$ and $\pi(i) = H_v - i + 1$ into (5) concludes the proof of Corollary 1. ■

We now consider the case $\widehat{M} = M$ in Theorem 2. Moreover, we focus on the case $h_1 \leq \frac{H_e}{M}$ and we show that, for this case, the upper bound on the secrecy capacity C_s is given by the right-hand side of (4) in Theorem 1.

Corollary 2. Assume that $\widehat{M} = M$ and $h_1 \leq \frac{H_e}{M}$ in Theorem 2. Then, it holds that

$$C_s \leq M \left(1 - \frac{K}{H_e}\right). \quad (8)$$

Proof: The proof is provided in Appendix B. ■

Remark 2. When $M = H_v$, the bound in Corollary 2 should match the one in Corollary 1 evaluated for $h_1 \leq \frac{H_e}{M}$. This is because from [10, Theorem 3] we know that the upper bound in Corollary 1 is the secrecy capacity when $H_v \leq M$. We now show that this is indeed the case. We start by noting that with $M = H_v$ and $h_1 \leq \frac{H_e}{M}$, we have that $h_i = \frac{H_e}{H_v}, i \in [H_v]$. This is because

$$H_e = \sum_{i=1}^{H_v} h_i \leq H_v h_1 \stackrel{h_1 = \max_{i \in [H_v]} h_i}{\leq} H_v h_1 \stackrel{h_1 \leq \frac{H_e}{M}}{=} M h_1 \leq H_e, \quad (9)$$

and hence, the only way to have the two inequalities above satisfied with equality is to have $h_i = \frac{H_e}{H_v}, i \in [H_v]$. Substituting this inside (7) with $M = H_v$ shows that the outer bounds in Corollary 2 and Corollary 1 match for $M = H_v$ and $h_1 \leq \frac{H_e}{M}$.

B. Lower Bound on C_s

We here derive a lower bound on the secrecy capacity C_s of an M -balanced 1-2-1 atomic network with $1 < M < H_v$. We start by providing the following lemma, which shows a characteristic of an M -balanced 1-2-1 atomic network.

Lemma 1. Consider an M -balanced 1-2-1 atomic network with $h_1 \geq h_2 \geq \dots \geq h_{H_v}$ (without loss of generality). Then, we can create H_e groups with the following two characteristics: (i) each group contains exactly M vertex-disjoint paths from S to D ; and (ii) each of the H_e paths from S to D appears in exactly M groups.

Proof: The proof of Lemma 1 is in Appendix C. ■

We now design a secure transmission scheme for an M -balanced 1-2-1 atomic network with $1 < M < H_v$ and with K eavesdropped edges which achieves the secrecy rate in Theorem 1. The scheme consists of four phases,

which are next described. The key generation and encoding phases are the same as those in [2]. The transmission occurs in $T = H_e$ rounds and it leverages the result in Lemma 1.

1) Key generation. We generate MK uniform random packets, denoted by $X_{[MK]}$, and create MH_e linear combinations of them by pre-multiplying $X_{[MK]}$ by a maximum distance separable (MDS) code matrix V of size $MH_e \times MK$, i.e.,

$$f(X) = VX_{[MK]}. \quad (10)$$

In what follows, we will refer to each row of $f(X)$ in (10) as a *key*. Note that the MH_e rows of $f(X)$ are linearly independent.

2) Encoding. We take $M(H_e - K)$ message packets $W_j, j \in [M(H_e - K)]$, and we encode them using $f(X)$ in (10). In particular, this encoding operation is as follows,

$$T_i = \begin{cases} f_i(X) & i \in [MK], \\ f_i(X) + W_{i-MK} & i \in [MK + 1 : MH_e], \end{cases} \quad (11)$$

where $f_i(X), i \in [MH_e]$, is the i -th row of $f(X)$ in (10).

3) Transmission. For the transmission phase, we start by creating H_e groups of paths that have the two properties stated in Lemma 1. This construction is provided in Algorithm 1, which returns $\mathcal{P}_M = \{\mathcal{P}^{(t)}\}_{t=1}^{H_e}$, where $\mathcal{P}^{(t)}$ is such that $|\mathcal{P}^{(t)}| = M, t \in [H_e]$. In other words, $\mathcal{P}_M$ contains H_e groups of paths that have the two properties stated in Lemma 1.

Then, the source S transmits the packets T_i 's in (11) over the paths $p \in \mathcal{P}^{(t)}, t \in [H_e]$. In particular, the transmission takes place in H_e rounds and, at each round, M packets are transmitted. For instance, for $t = 1$, the source S transmits M packets simultaneously, one through each of the M vertex-disjoint paths in $\mathcal{P}^{(1)}$. Note that this is possible since: (i) at each point in time, S can simultaneously transmit to M nodes, and (ii) the M paths in $\mathcal{P}^{(t)}$ are vertex-disjoint. Thus, the 1-2-1 constraints are satisfied. At the end of round $t = 1$, S has transmitted $T_i, i \in [M]$. After this, round $t = 2$ starts and S transmits M packets simultaneously (i.e., in one network use), one through each of the M paths in $\mathcal{P}^{(2)}$. Then, round $t = 3$ will start and so on until round $t = H_e$. Thus, in each round $t \in [1 : H_e]$, M packets are sent by S, i.e., in H_e rounds a total of MH_e packets are transmitted by S.

4) Decoding. At the destination D (which, from the 1-2-1 constraints, can receive simultaneously from M nodes), the decoding is done by first finding the MK random packets $X_{[MK]}$ and then reconstructing the keys $f(X)$. Specifically, since the first MK received packets are just keys without messages (see (11)), the random packets $X_{[MK]}$ can be obtained as follows,

$$X_{[MK]} = (V_{[MK],[MK]})^{-1}T_{[MK]}, \quad (12)$$

where $V_{[MK],[MK]}$ is the sub-matrix of V obtained by just retaining the first MK rows and the first MK columns of V . Then, D can generate the keys $f(X)$ using $X_{[MK]}$ in (12) similar to (10). Finally, D decodes the messages $W_j, j \in [M(H_e - K)]$ as follows,

$$\widehat{W}_{i-MK} = T_i - f_i(X), \quad (13)$$

Algorithm 1 Construction of $\mathcal{P}_M = \{\mathcal{P}^{(t)}\}_{t=1}^{H_e}$

Input: $\mathbf{h} = [h_1, h_2, \dots, h_{H_v}]$, M , H_e .

- 1: Let G be the extended network, where each atomic subgraph has Mh_i edge-disjoint paths with $i \in [H_v]$.
 - 2: Set $t = 1$. Initialize $G^{(1)} = G$ and $\mathcal{P}_M = \emptyset$.
 - 3: **while** it is possible to select M atomic subgraphs of $G^{(t)}$ with non-zero edge-disjoint paths **do**
 - 4: Select M atomic subgraphs of $G^{(t)}$ that have the largest number of edge-disjoint paths.
 - 5: Select one path from each selected atomic subgraph and let $\mathcal{Q}$ denote them. Note that $|\mathcal{Q}| = M$.
 - 6: Set $\mathcal{P}^{(t)} = \mathcal{Q}$.
 - 7: Update $\mathcal{P}_M = \mathcal{P}_M \cup \mathcal{P}^{(t)}$.
 - 8: Update $G^{(t+1)} = G^{(t)} \setminus \mathcal{Q}$.
 - 9: Increment $t \leftarrow t + 1$.
 - 10: **end while**
 - 11: **return** $\mathcal{P}_M$.
-

where $i \in [MK + 1 : MH_e]$.

Security. In each network use, the adversary can receive a packet passing through an eavesdropped edge if the eavesdropped edge belongs to the paths used in that particular network use. Since the K eavesdropped edges can at most be part of K paths, and each of the H_e edge-disjoint paths is used exactly M times (because of the property in Lemma 1), then the adversary will eavesdrop at most MK packets, which are linearly independent thanks to the property of MDS codes (see (11)). Thus, the scheme securely transmits a total of $M(H_e - K)$ message packets in H_e network uses. This implies that a rate $R = M \left(1 - \frac{K}{H_e}\right)$ can be securely achieved and hence, the secrecy capacity C_s in Theorem 1 is achievable.

We now conclude this section with an example that shows how the proposed transmission scheme operates.

Example 4. Consider the 1-2-1 atomic network $\mathbf{h} = [3, 2, 2]$ in Figure 1 with $M = 2$. Assume $K = 3$. Then, the proposed transmission scheme operates as follows:

- 1) We generate $MK = 6$ uniformly random packets $X_{[6]}$ and extend them to $MH_e = 14$ to form $f_i(X), i \in [14]$ using an MDS code matrix of size 14×6 .
- 2) We encode $M(H_e - K) = 8$ messages $W_i, i \in [8]$ as,

$$T_i = \begin{cases} f_i(X) & i \in [6], \\ f_i(X) + W_{i-6} & i \in [7 : 14]. \end{cases} \quad (14)$$

- 3) From Algorithm 1 we obtain

$$\mathcal{P}_M = \left\{ \mathcal{P}^{(t)} : t \in [7] \right\}, \quad (15a)$$

where

$$\begin{aligned}
\mathcal{P}^{(1)} &= \{p_1, p_7\}, & \mathcal{P}^{(2)} &= \{p_2, p_{11}\}, \\
\mathcal{P}^{(3)} &= \{p_3, p_{12}\}, & \mathcal{P}^{(4)} &= \{p_4, p_8\}, \\
\mathcal{P}^{(5)} &= \{p_5, p_9\}, & \mathcal{P}^{(6)} &= \{p_{10}, p_{13}\}, \\
\mathcal{P}^{(7)} &= \{p_6, p_{14}\},
\end{aligned} \tag{15b}$$

where $\{p_i\}_{i=1}^6$ is the set of $Mh_1 = 6$ paths in the first extended subgraph, $\{p_i\}_{i=7}^{10}$ is the set of $Mh_2 = 4$ paths in the second extended subgraph, and $\{p_i\}_{i=11}^{14}$ is the set of $Mh_3 = 4$ paths in the third extended subgraph. For transmission, we use the network $H_e = 7$ times: (a) the $M = 2$ paths in each $\mathcal{P}^{(t)}$, $t \in [7]$ can be used to simultaneously transmit $M = 2$ packets, e.g., p_1 and p_7 can be used to transmit T_1 and T_2 in the first network use; and (b) each path is used exactly $M = 2$ times.

- 4) Upon receiving $T_i, i \in [14]$, D recovers $W_j, j \in [8]$ using the property of the MDS code matrix (see (12) and (13)).

The adversary can eavesdrop at most $MK = 6$ packets, which are encoded with independent keys. Thus, the eavesdropper cannot learn anything about $W_j, j \in [8]$ and hence, we obtain a secrecy rate $R = \frac{14-6}{7} = \frac{8}{7}$.

IV. CONCLUSION

In this work, we have studied the secrecy capacity of 1-2-1 atomic networks - first introduced in [10] - which capture the directional nature of mmWave communication. Building on our earlier results in [10], we have established new lower and upper bounds on the secrecy capacity of 1-2-1 atomic networks, closing a previously unresolved regime in which the atoms have comparable workloads - a setting that is particularly relevant for symmetric network topologies. In particular, our novel upper bound is tighter than the bound we derived in [10]. Moreover, our new secure transmission scheme achieves a larger secrecy-communication rate than the previously proposed scheme in [10]. Lastly, we have shown that the upper bound and the achievable secrecy rate match for M -balanced 1-2-1 atomic networks with $M < H_v$, thereby fully characterizing the secrecy capacity in this regime. Together with the results in [10], our findings in this paper complete the characterization of the secrecy capacity of M -balanced 1-2-1 atomic networks.

APPENDIX A

PROOF OF THE UPPER BOUND OF C_s IN THEOREM 2

Let $\mathcal{E}_{D-} \subset \mathcal{E}$ be the set of all incoming edges to the destination D. At any time instant t , the destination can receive packets from at most M incoming edges among $\mathcal{E}_{D-}$, which we denote by $\mathcal{D}_t \subseteq \mathcal{E}_{D-}$ with $|\mathcal{D}_t| \leq M$. We define $\mathcal{V}_t = \{v_i \in \mathcal{V}_i : i \in [H_v], \mathcal{E}_i \cap \mathcal{D}_t \neq \emptyset\}$ with $|\mathcal{V}_t| \leq \widehat{M}$ (recall that $\widehat{M} := \min(M, H_v)$). Now, by starting from Lemma 3 in Appendix D, we have that

$$\begin{aligned}
nR &\stackrel{(a)}{\leq} \varepsilon + \sum_{t=1}^n \sum_{i: v_i \in \mathcal{V}_t} \frac{h_i - K_i}{h_i} \\
&\stackrel{(b)}{\leq} \varepsilon + \min_{\substack{K_i \in \mathbb{N} \cup \{0\}, i \in [H_v]: \\ \sum_{i=1}^{H_v} K_i = K, \\ 0 \leq K_i \leq h_i, i \in [H_v]}} \sum_{t=1}^n \sum_{i: v_i \in \mathcal{V}_t} \frac{h_i - K_i}{h_i}
\end{aligned}$$

$$\begin{aligned}
& \stackrel{(c)}{\leq} \varepsilon + \max_{\substack{\mathcal{V}_1, \mathcal{V}_2, \dots, \mathcal{V}_n: \\ |\mathcal{V}_t| \leq \widehat{M}, \forall t \in [n]}} \min_{\substack{K_i \in \mathbb{N} \cup \{0\}, i \in [H_v]: \\ \sum_{i=1}^{H_v} K_i = K, \\ 0 \leq K_i \leq h_i, i \in [H_v]}} \sum_{t=1}^n \sum_{i: v_i \in \mathcal{V}_t} \frac{h_i - K_i}{h_i} \\
& \stackrel{(d)}{=} \varepsilon + \max_{\substack{y_{ti} \in \{0,1\}, t \in [n], i \in [H_v]: \\ \sum_{i=1}^{H_v} y_{ti} \leq \widehat{M}, \forall t \in [n]}} \min_{\substack{K_j \in \mathbb{N} \cup \{0\}, j \in [H_v]: \\ \sum_{i=1}^{H_v} K_i = K \\ 0 \leq K_j \leq h_j, j \in [H_v]}} \sum_{t=1}^n \sum_{i=1}^{H_v} y_{ti} \left(\frac{h_i - K_i}{h_i} \right) \\
& \stackrel{(e)}{=} \varepsilon + \max_{\substack{a_i \in \mathbb{N} \cup \{0\}, i \in [H_v]: \\ 0 \leq a_i \leq n \\ \sum_{i=1}^{H_v} a_i \leq n \widehat{M}}} \min_{\substack{K_i \in \mathbb{N} \cup \{0\}, i \in [H_v]: \\ \sum_{i=1}^{H_v} K_i = K \\ 0 \leq K_i \leq h_i, i \in [H_v]}} \sum_{i=1}^{H_v} a_i \left(1 - \frac{K_i}{h_i} \right) \\
& \stackrel{(f)}{=} \varepsilon + \max_{\mathbf{a} \in \mathcal{A}_n} \min_{\mathbf{K} \in \mathcal{K}} \left(\sum_{i=1}^{H_v} a_i - \sum_{i=1}^{H_v} \frac{a_i}{h_i} K_i \right), \tag{16}
\end{aligned}$$

where the labeled (in)equalities follow from: (a) using Lemma 3 in Appendix D, where $K_i \in [0 : h_i], i \in [H_v]$ such that $\sum_{i=1}^{H_v} K_i = K$; (b) the fact that the upper bound in (a) holds for any $\mathbf{K} = (K_1, \dots, K_{H_v})$ with $\sum_{i=1}^{H_v} K_i = K$ and hence, it must hold also for the tuple which minimizes the upper bound in (a); (c) the fact that at any time instant $t \in [n]$, the atoms in $\mathcal{V}_t$ are the only active atoms which contribute in transmitting packets from the source to the destination and hence, by maximizing over all sets $\mathcal{V}_t$ for $t \in [n]$, we get an upper bound on the minimization problem in (b) in the form of a max-min optimization problem; (d) defining a binary decision variable $y_{ti} = \mathbb{1}_{\mathcal{V}_t}(v_i)$, for all $t \in [n]$ and $i \in [H_v]$, which converts the optimization problem to an integer program; (e) further simplifying the integer programming by defining $a_i := \sum_{t=1}^n y_{ti}$ for all $i \in [H_v]$; in particular, $y_{ti} = 1$ if $v_i \in \mathcal{V}_t$ and $y_{ti} = 0$ otherwise; and (f) letting $\mathcal{A}_n = \left\{ \mathbf{a} \in [0 : n]^{H_v} : \sum_{i=1}^{H_v} a_i \leq n \widehat{M} \right\}$, and $\mathcal{K} = \left\{ \mathbf{K} : K_i \in \mathbb{N} \cup \{0\}, i \in [H_v], 0 \leq K_i \leq h_i, \sum_{i=1}^{H_v} K_i = K \right\}$.

Dividing both sides of (16) by n , we arrive at

$$R \leq \frac{\varepsilon}{n} + \frac{1}{n} \max_{\mathbf{a} \in \mathcal{A}_n} \min_{\mathbf{K} \in \mathcal{K}} \left(\sum_{i=1}^{H_v} a_i - \sum_{i=1}^{H_v} \frac{a_i}{h_i} K_i \right), \tag{17}$$

and hence, in the limit $n \rightarrow \infty$, the secure capacity C_s can be upper bounded as follows,

$$\begin{aligned}
C_s & \leq \lim_{n \rightarrow \infty} \frac{1}{n} \max_{\mathbf{a} \in \mathcal{A}_n} \min_{\mathbf{K} \in \mathcal{K}} \left(\sum_{i=1}^{H_v} a_i - \sum_{i=1}^{H_v} \frac{a_i}{h_i} K_i \right) \\
& \stackrel{(a)}{\leq} \max_{\alpha \in \mathcal{A}'} \min_{\mathbf{K} \in \mathcal{K}} \sum_{i=1}^{H_v} \frac{\alpha_i}{h_i} (h_i - K_i) \\
& \stackrel{(b)}{=} \max_{\alpha \in \cup_{\tau \in (0, \widehat{M}]} \mathcal{A}_\tau} \min_{\mathbf{K} \in \mathcal{K}} \sum_{i=1}^{H_v} \frac{\alpha_i}{h_i} (h_i - K_i) \\
& \stackrel{(c)}{=} \max_{\alpha \in \mathcal{A}_{\widehat{M}}} \min_{\mathbf{K} \in \mathcal{K}} \sum_{i=1}^{H_v} \frac{\alpha_i}{h_i} (h_i - K_i) \\
& \stackrel{(d)}{=} \max_{\alpha \in \mathcal{A}_{\widehat{M}}} \sum_{i=1}^{H_v} \alpha_{\pi(i)} \left(1 - \frac{K_{\pi(i)}^*}{h_{\pi(i)}} \right) \tag{18}
\end{aligned}$$

where the labeled (in)equalities follow from: (a) the fact that we can upper bound the problem by the change of variable $\alpha_i = \frac{a_i}{n}$, $i \in [H_v]$, and enlarging the feasible region $\mathcal{A}_n$ to $\mathcal{A}' = \left\{ \alpha \in [0, 1]^{H_v} : \sum_{i=1}^{H_v} \alpha_i \leq \widehat{M} \right\}$; (b)

since $\mathcal{A}' = \cup_{\tau \in (0, \widehat{M})} \mathcal{A}_\tau$, where $\mathcal{A}_\tau = \left\{ \alpha \in [0, 1]^{H_v} : \sum_{i=1}^{H_v} \alpha_i = \tau \right\}$ for all $\tau \in (0, \widehat{M})$; (c) the result in Lemma 4 in Appendix E; and (d) the result in Lemma 5 in Appendix F where

$$\pi(i) = \left\{ j \in [H_v] : \frac{\alpha_j}{h_j} \text{ is the } i\text{-th maximum element in the vector } \alpha \oslash \mathbf{h} \right\}, \quad \forall i \in [H_v], \quad (19)$$

and

$$K_{\pi(i)}^* = \min \left\{ K - \sum_{j=1}^{i-1} K_{\pi(j)}^*, h_{\pi(i)} \right\}, \quad i \in [H_v]. \quad (20)$$

This concludes the proof of Theorem 2.

APPENDIX B

PROOF OF COROLLARY 2

The proof of Corollary 2 consists of solving the optimization problem in (5) when $\widehat{M} = M$ and $h_1 \leq \frac{H_e}{M}$. We start by noting that the optimization problem in (5) can be rewritten as follows,

$$\begin{aligned} \max_{\alpha \in \mathcal{A}_{\widehat{M}}} \sum_{i=1}^{H_v} \alpha_{\pi(i)} \left(1 - \frac{K_{\pi(i)}^*}{h_{\pi(i)}} \right) &\stackrel{(a)}{=} \max_{\pi \in \mathcal{P}_{H_v}} \max_{\beta \in \mathcal{B}_\pi} \sum_{i=1}^{H_v} \beta_{\pi(i)} \left(1 - \frac{K_{\pi(i)}^*}{h_{\pi(i)}} \right) \\ &\stackrel{(b)}{=} \max_{\pi \in \mathcal{P}_{H_v}} \max_{\beta \in \mathcal{B}_\pi} \sum_{i=1}^{H_v} \beta_i \left(1 - \frac{K_i^\pi}{h_i} \right), \end{aligned} \quad (21)$$

where the first equality follows by letting $\mathcal{P}_{H_v}$ be the set of all the $H_v!$ permutations of $[H_v]$ and

$$\mathcal{B}_\pi = \left\{ \beta \in [0, 1]^{H_v} : \sum_{k=1}^{H_v} \beta_k = M, \frac{\beta_{\pi(i)}}{h_{\pi(i)}} \geq \frac{\beta_{\pi(j)}}{h_{\pi(j)}}, \forall i < j \right\}. \quad (22)$$

and in (b) we let $K_j^\pi := K_{\pi(i)}^*$ where $\pi(i) = j$; we use the notation K_j^π to highlight that the values of $K_{\pi(i)}^*$'s depend on the permutation π (see Theorem 2). For each $\pi \in \mathcal{P}_{H_v}$, the coefficients $\left(1 - \frac{K_i^\pi}{h_i} \right), i \in [H_v]$ are well-defined and fixed. Thus, the inner optimization problem of (21), that is,

$$P_{\text{inner}}(\pi) := \max_{\beta \in \mathcal{B}_\pi} \sum_{i=1}^{H_v} \beta_i \left(1 - \frac{K_i^\pi}{h_i} \right), \quad (23)$$

is a linear program in β . Since for each $\pi \in \mathcal{P}_{H_v}$, the feasible set $\mathcal{B}_\pi$ is non-empty, the problem $P_{\text{inner}}(\pi)$ is feasible. Hence, strong duality holds. We proceed by constructing the Lagrangian associated with $P_{\text{inner}}(\pi)$, denoted as $\mathcal{L}_\pi : \mathbb{R}^{H_v} \times \mathbb{R} \times \mathbb{R}^{H_v-1} \times \mathbb{R}^{H_v} \times \mathbb{R}^{H_v} \rightarrow \mathbb{R}$, and defined as

$$\begin{aligned} \mathcal{L}_\pi(\beta, \lambda, \mu, \eta, \gamma) &:= - \sum_{i=1}^{H_v} \beta_i \left(1 - \frac{K_i^\pi}{h_i} \right) + \lambda \left(\sum_{i=1}^{H_v} \beta_i - M \right) + \sum_{i=1}^{H_v-1} \mu_i \left(\frac{\beta_{\pi(i+1)}}{h_{\pi(i+1)}} - \frac{\beta_{\pi(i)}}{h_{\pi(i)}} \right) \\ &\quad + \sum_{i=1}^{H_v} \eta_i (\beta_i - 1) - \sum_{i=1}^{H_v} \gamma_i \beta_i, \end{aligned} \quad (24)$$

where $(\lambda, \mu, \eta, \gamma)$ are the Lagrange multipliers (the dual variables) associated with the constraints defining $\mathcal{B}_\pi$ in (23). By strong duality, a point β^* is an optimal solution if and only if there exist dual variables $(\lambda^*, \mu^*, \eta^*, \gamma^*)$ that, together with β^* , satisfy the Karush-Kuhn-Tucker (KKT) conditions. Let $\pi \in \mathcal{P}_{H_v}$ be an arbitrary permutation and define $\omega_i^\pi := \left(1 - \frac{K_i^\pi}{h_i} \right), i \in [H_v]$. The KKT conditions for $P_{\text{inner}}(\pi)$ in (23) are given by:

- 1) *Stationarity*: To facilitate the differentiation of the Lagrangian function $\mathcal{L}_\pi$ in (24) with respect to β_i , we introduce the inverse permutation π^{-1} of π , defined by $\pi^{-1}(\pi(i)) = i$ for all $i \in [H_v]$. Then, the third summation of $\mathcal{L}_\pi$ in (24) can be rewritten as

$$\begin{aligned} \sum_{i=1}^{H_v-1} \mu_i \left(\frac{\beta_{\pi(i+1)}}{h_{\pi(i+1)}} - \frac{\beta_{\pi(i)}}{h_{\pi(i)}} \right) &= \sum_{i=2}^{H_v} \mu_{i-1} \frac{\beta_{\pi(i)}}{h_{\pi(i)}} - \sum_{j=1}^{H_v-1} \mu_j \frac{\beta_{\pi(j)}}{h_{\pi(j)}} \\ &\stackrel{(a)}{=} \sum_{i=1}^{H_v} \mu_{i-1} \frac{\beta_{\pi(i)}}{h_{\pi(i)}} - \sum_{j=1}^{H_v} \mu_j \frac{\beta_{\pi(j)}}{h_{\pi(j)}} \\ &\stackrel{(b)}{=} \sum_{i=1}^{H_v} \mu_{\pi^{-1}(i)-1} \frac{\beta_i}{h_i} - \sum_{j=1}^{H_v} \mu_{\pi^{-1}(j)} \frac{\beta_j}{h_j}, \end{aligned} \quad (25)$$

where (a) follows by letting $\mu_0 = \mu_{H_v} = 0$, and (b) follows from using the inverse permutation π^{-1} . Plugging (25) into (24) and differentiating $\mathcal{L}_\pi$ in (24) with respect to β_i , we obtain the following stationarity condition:

$$\frac{\partial}{\partial \beta_i} \mathcal{L}_\pi(\beta^*, \lambda^*, \mu^*, \eta^*, \gamma^*) = -\omega_i^\pi + \lambda^* + \frac{\mu_{\pi^{-1}(i)-1}^*}{h_i} - \frac{\mu_{\pi^{-1}(i)}^*}{h_i} + \eta_i^* - \gamma_i^* = 0, \quad i \in [H_v], \quad (26)$$

with $\mu_0^* = \mu_{H_v}^* = 0$.

- 2) *Primal feasibility*:

$$\beta^* \in \mathcal{B}_\pi, \quad (27)$$

where

$$\mathcal{B}_\pi = \left\{ \beta \in [0, 1]^{H_v} : \sum_{i=1}^{H_v} \beta_i = M, \frac{\beta_{\pi(1)}}{h_{\pi(1)}} \geq \frac{\beta_{\pi(2)}}{h_{\pi(2)}} \geq \dots \geq \frac{\beta_{\pi(H_v)}}{h_{\pi(H_v)}} \right\}. \quad (28)$$

- 3) *Dual feasibility*:

$$\lambda^* \in \mathbb{R},$$

$$\mu_i^* \geq 0, \quad \forall i \in [H_v - 1], \quad (29)$$

$$\eta_i^* \geq 0, \quad \forall i \in [H_v],$$

$$\gamma_i^* \geq 0, \quad \forall i \in [H_v].$$

- 4) *Complementary slackness*:

$$\begin{aligned} \mu_i^* \left(\frac{\beta_{\pi(i+1)}^*}{h_{\pi(i+1)}} - \frac{\beta_{\pi(i)}^*}{h_{\pi(i)}} \right) &= 0, \quad \forall i \in [H_v - 1], \\ \eta_i^* (\beta_i^* - 1) &= 0, \quad \forall i \in [H_v], \\ \gamma_i^* \beta_i^* &= 0, \quad \forall i \in [H_v]. \end{aligned} \quad (30)$$

The following lemma provides an optimal solution β^* for $\mathbf{P}_{\text{inner}}(\pi)$ in (23) when $h_1 \leq \frac{H_e}{M}$.

Lemma 2. Assume $h_1 \leq \frac{H_e}{M}$. Then, for any $\pi \in \mathcal{P}_{H_v}$ an optimal solution β^* for $\mathbf{P}_{\text{inner}}(\pi)$ in (23) is given by

$$\beta_i^* = \frac{M h_i}{H_e}, \quad \forall i \in [H_v]. \quad (31)$$

Proof: The proof is provided in Appendix G. ■

Substituting β_i^* in Lemma 2 inside (21), we obtain an optimal solution for the optimization problem in (5), i.e.,

$$\begin{aligned}
C_s &\leq \max_{\pi \in \mathcal{P}_{H_v}} \sum_{i=1}^{H_v} \frac{M h_{\pi(i)}}{H_e} \left(1 - \frac{K_{\pi(i)}^*}{h_{\pi(i)}} \right) \\
&= \max_{\pi \in \mathcal{P}_{H_v}} \frac{M}{H_e} \left(\sum_{i=1}^{H_v} h_{\pi(i)} - \sum_{i=1}^{H_v} K_{\pi(i)}^* \right) \\
&= \frac{M}{H_e} (H_e - K) \\
&= M \left(1 - \frac{K}{H_e} \right). \tag{32}
\end{aligned}$$

This concludes the proof of Corollary 2.

APPENDIX C

PROOF OF LEMMA 1

The proof leverages [10, Lemma 1], which showed that in a 1-2-1 atomic network with $h_1 \geq h_2 \geq \dots \geq h_{H_v}$ there are at least $P_\ell(h_{[H_v]})$ groups of ℓ vertex-disjoint paths, where

$$P_1(h_{[H_v]}) = \sum_{i=1}^{H_v} h_i, \tag{33a}$$

and, for all $\ell \in [2 : H_v]$, it holds that

$$P_\ell(h_{[H_v]}) = \begin{cases} P_{\ell-1}(h_{[2:H_v]}) & \text{if } h_1 \geq P_{\ell-1}(h_{[2:H_v]}), \\ \left\lfloor \frac{\sum_{i=1}^{H_v} h_i}{\ell} \right\rfloor & \text{otherwise.} \end{cases} \tag{33b}$$

Moreover, using the algorithm proposed in [10, Lemma 1], each of the H_e paths either appears in exactly one group or does not appear in any group. The number of paths that do *not* appear in any group is given by

$$\bar{N} = \sum_{i=1}^{H_v} h_i - \ell \left\lfloor \frac{\sum_{i=1}^{H_v} h_i}{\ell} \right\rfloor. \tag{34}$$

Proving Lemma 1 is therefore equivalent to proving that, when $h_1 \leq \frac{H_e}{M}$, we have that $P_M(Mh_{[H_v]}) = H_e$. This is because we want H_e groups such that: (i) each group contains exactly M vertex-disjoint paths from S to D and hence, $\ell = M$ in (33); and (ii) each of the H_e paths from S to D appears in exactly M groups and hence, we replace h_i with Mh_i , $i \in [H_v]$ in (33), which from (34) leads to $\bar{N} = 0$. Thus, we now prove that $P_M(Mh_{[H_v]}) = H_e$. Toward this end, we start by proving the following result. For $i \in [1 : M - 1]$, assume that

$$P_i(Mh_{[M-i+1:H_v]}) = \left\lfloor \frac{M \sum_{j=M-i+1}^{H_v} h_j}{i} \right\rfloor, \tag{35}$$

then, it holds that $Mh_{M-i} \leq P_i(Mh_{[M-i+1:H_v]})$. The proof is by induction. Note that for the base case $i = 1$, the assumption in (35) holds true. By using $i = 1$ inside (35), we indeed obtain $P_1(Mh_{[M:H_v]}) = M \sum_{j=M}^{H_v} h_j$, which is equal to (33a). Now, consider some $i \in [1 : M - 1]$ and assume that (35) holds. Our goal is to show that $Mh_{M-i} \leq P_i(Mh_{[M-i+1:H_v]})$. We rewrite M as $M = ic + t$, where $c \in \mathbb{N}$ and $t \in [0 : i - 1]$. By using this inside (35), we obtain

$$P_i(Mh_{[M-i+1:H_v]}) = \left\lfloor \frac{M \sum_{j=M-i+1}^{H_v} h_j}{i} \right\rfloor$$

$$\begin{aligned}
&= \left\lfloor \frac{(ic + t) \sum_{j=M-i+1}^{H_v} h_j}{i} \right\rfloor \\
&= c \sum_{j=M-i+1}^{H_v} h_j + \left\lfloor \frac{t}{i} \sum_{j=M-i+1}^{H_v} h_j \right\rfloor \\
&= c \left(H_e - \sum_{j=1}^{M-i} h_j \right) + \left\lfloor \frac{t}{i} \left(H_e - \sum_{j=1}^{M-i} h_j \right) \right\rfloor \\
&\stackrel{(a)}{\geq} c (H_e - (M-i)h_1) + \left\lfloor \frac{t}{i} (H_e - (M-i)h_1) \right\rfloor \\
&\stackrel{(b)}{\geq} ci h_1 + \left\lfloor \frac{t}{i} (i h_1) \right\rfloor \\
&= h_1 (ci + t) = M h_1 \stackrel{(a)}{\geq} M h_{M-i},
\end{aligned} \tag{36}$$

where (a) follows since $h_1 \geq h_2 \geq \dots \geq h_{H_v}$ and (b) is due to the fact that $h_1 \leq \frac{H_e}{M}$. Thus, we have proved that, for $i \in [1 : M-1]$, under the assumption in (35), it holds that $M h_{M-i} \leq P_i (M h_{[M-i+1:H_v]})$. The proof of Lemma 1 is concluded by setting $i = M-1$, which leads to $M h_1 \leq P_{M-1} (M h_{[2:H_v]})$. Using this inside (33b) leads to

$$P_M (M h_{[H_v]}) = \left\lfloor \frac{\sum_{i=1}^{H_v} M h_i}{M} \right\rfloor = H_e. \tag{37}$$

This concluded the proof of Lemma 1.

APPENDIX D

PROOF OF (a) IN (16)

Lemma 3. Consider a 1-2-1 atomic network with underlying graph G described by $\mathbf{h} = [h_1, \dots, h_{H_v}]$. Then, for the source S to communicate with the destination D with zero error in n network uses, while satisfying the secrecy constraint in (1), it must hold that

$$nR \leq \varepsilon + \sum_{t=1}^n \sum_{i: v_i \in \mathcal{V}_t} \frac{h_i - K_i}{h_i},$$

where $\mathcal{V}_t = \{v_i \in \mathcal{V}_i : i \in [H_v], \mathcal{E}_i \cap \mathcal{D}_t \neq \emptyset\}$ with $\mathcal{D}_t$ being the set of incoming edges to D from which D receives information at time t , and $K_i \in [0 : h_i], i \in [H_v]$ such that $\sum_{i=1}^{H_v} K_i = K$.

Proof: Let $\mathcal{E}_{D-} \subset \mathcal{E}$ be the set of all incoming edges to the destination D . We also let $T_{\mathcal{E}_{D-}}^t = \{T_e^t : e \in \mathcal{E}_{D-}\}$ and $T_{\mathcal{E}_{D-}}^{[n]}$ denote the packets received by the destination D through the incoming edges in $\mathcal{E}_{D-}$ at time instance $t \in [n]$ and over all n network uses, respectively. At any time instant t , the destination can receive packets $T_{\mathcal{E}_{D-}}^t$ from at most M incoming edges among $\mathcal{E}_{D-}$, which we denote by $\mathcal{D}_t \subseteq \mathcal{E}_{D-}$ with $|\mathcal{D}_t| \leq M$. We define $\mathcal{V}_t = \{v_i \in \mathcal{V}_i : i \in [H_v], \mathcal{E}_i \cap \mathcal{D}_t \neq \emptyset\}$ with $|\mathcal{V}_t| \leq \widehat{M}$ (recall that $\widehat{M} := \min(M, H_v)$), and we denote the set of incoming edges in $\mathcal{V}_t$ by $\mathcal{E}_{\mathcal{V}_t}$. Now we have that

$$\begin{aligned}
nR &\stackrel{(a)}{=} H(W) - H(W | T_{\mathcal{E}_{D-}}^{[n]}) \\
&= I\left(W; \bigcup_{t=1}^n T_{\mathcal{E}_{D-}}^t\right)
\end{aligned}$$

$$\begin{aligned}
& \stackrel{(b)}{=} I \left(W; \bigcup_{t=1}^n T_{\mathcal{D}_t}^t \right) \\
& \leq I \left(W; \bigcup_{t=1}^n T_{\mathcal{D}_t}^t, \bigcup_{t=1}^n T_{\mathcal{E}_{\mathcal{V}_t^-}}^t \right) \\
& \stackrel{(c)}{=} I \left(W; \bigcup_{t=1}^n T_{\mathcal{E}_{\mathcal{V}_t^-}}^t \right) \\
& \stackrel{(d)}{\leq} \varepsilon + I \left(W; \bigcup_{t=1}^n T_{\mathcal{E}_{\mathcal{V}_t^-} \setminus \mathcal{Z}}^t \middle| \bigcup_{t=1}^n T_{\mathcal{Z}}^t \right) \\
& \stackrel{(e)}{\leq} \varepsilon + H \left(\bigcup_{t=1}^n T_{\mathcal{E}_{\mathcal{V}_t^-} \setminus \mathcal{Z}}^t \middle| \bigcup_{t=1}^n T_{\mathcal{Z}}^t \right) \\
& \stackrel{(f)}{\leq} \varepsilon + \sum_{t=1}^n H \left(T_{\mathcal{E}_{\mathcal{V}_t^-} \setminus \mathcal{Z}}^t \middle| T_{\mathcal{Z}}^t \right) \\
& \stackrel{(g)}{=} \varepsilon + \sum_{t=1}^n H \left(\bigcup_{i: v_i \in \mathcal{V}_t} T_{\mathcal{E}_{v_i^-} \setminus \mathcal{Z}_i}^t \middle| T_{\mathcal{Z}}^t \right) \\
& \stackrel{(h)}{\leq} \varepsilon + \sum_{t=1}^n \sum_{i: v_i \in \mathcal{V}_t} H \left(T_{\mathcal{E}_{v_i^-} \setminus \mathcal{Z}_i}^t \middle| T_{\mathcal{Z}_i}^t \right) \\
& \stackrel{(i)}{\leq} \varepsilon + \sum_{t=1}^n \sum_{i: v_i \in \mathcal{V}_t} \frac{h_i - K_i}{h_i}, \tag{38}
\end{aligned}$$

where the labeled (in)equalities follow from: (a) the constraint for reliable decoding; (b) the fact that the destination, at time t , receives only through the edges in $\mathcal{D}_t$ (and no information is received through the edges in $\mathcal{E}_{\mathcal{D}^-} \setminus \mathcal{D}_t$); (c) the data processing inequality; (d) the secrecy constraint in (1) and the fact that the eavesdropper does not move over time, i.e., the set $\mathcal{Z}$ is fixed for all $t \in [n]$; (e) the fact that the entropy of a discrete random variable is non-negative; (f) using the chain rule for the entropy and the fact that conditioning does not increase the entropy; (g) letting $\mathcal{E}_{v_i^-}$ be the set of incoming edges in the atom v_i and defining, for the atom v_i , the set of the eavesdropped edges $\mathcal{Z}_i \subseteq \mathcal{E}_i$ with $|\mathcal{Z}_i| = K_i$ such that $\sum_{i=1}^{H_v} K_i = |\mathcal{Z}| = K$; (h) using the chain rule for the entropy and the fact that conditioning does not increase the entropy; and (i) using Lemma 1 in [2] and noticing that $K_i \leq h_i$, i.e., the eavesdropper cannot eavesdrop more than the available edges. This concludes the proof of Lemma 3. $\blacksquare$

APPENDIX E

PROOF OF (c) IN (18)

Lemma 4. For any $\tau \in \mathbb{R}^+$, define $\mathcal{A}_\tau = \left\{ \boldsymbol{\alpha} \in [0, 1]^{H_v} : \sum_{i=1}^{H_v} \alpha_i = \tau \right\}$. If $\tau_1 \geq \tau_2$, then

$$\max_{\boldsymbol{\alpha} \in \mathcal{A}_{\tau_1}} \min_{\mathbf{K} \in \mathcal{K}} \sum_{i=1}^{H_v} \frac{\alpha_i}{h_i} (h_i - K_i) \geq \max_{\boldsymbol{\alpha} \in \mathcal{A}_{\tau_2}} \min_{\mathbf{K} \in \mathcal{K}} \sum_{i=1}^{H_v} \frac{\alpha_i}{h_i} (h_i - K_i), \tag{39}$$

where $\mathcal{K} = \left\{ \mathbf{K} : K_i \in \mathbb{N} \cup \{0\}, i \in [H_v] : 0 \leq K_i \leq h_i, \sum_{i=1}^{H_v} K_i = K \right\}$.

Proof: Let $\tau_1 = \tau$ and $\tau_2 = \tau - c$, where c is such that $0 < c < \tau$. Note that, from the definition of $\mathcal{A}_\tau$, we need $\tau \leq H_v$. With this, we have that

$$\begin{aligned}
\max_{\mathbf{b} \in \mathcal{A}_{\tau-c}} \min_{\mathbf{K} \in \mathcal{K}} \sum_{i=1}^{H_v} \frac{b_i}{h_i} (h_i - K_i) &\stackrel{(a)}{=} \min_{\mathbf{K} \in \mathcal{K}} \sum_{i=1}^{H_v} \frac{b_i^*}{h_i} (h_i - K_i) \\
&\stackrel{(b)}{\leq} \min_{\mathbf{K} \in \mathcal{K}} \sum_{i=1}^{H_v} \frac{b_i^* + \varepsilon_i^*}{h_i} (h_i - K_i) \\
&\stackrel{(c)}{=} \min_{\mathbf{K} \in \mathcal{K}} \sum_{i=1}^{H_v} \frac{\alpha_i^*}{h_i} (h_i - K_i) \\
&\stackrel{(d)}{\leq} \max_{\boldsymbol{\alpha} \in \mathcal{A}_\tau} \min_{\mathbf{K} \in \mathcal{K}} \sum_{i=1}^{H_v} \frac{\alpha_i}{h_i} (h_i - K_i), \tag{40}
\end{aligned}$$

where the labeled (in)equalities follow from: (a) letting $\mathbf{b}^* = \operatorname{argmax}_{\mathbf{b} \in \mathcal{A}_{\tau-c}} \min_{\mathbf{K} \in \mathcal{K}} \sum_{i=1}^{H_v} \frac{b_i}{h_i} (h_i - K_i)$; (b) adding ε^* , where the non-negative ε_i^* for all $i \in [H_v]$ is built as follows: let $t \in [H_v]$ be such that $\sum_{i=1}^{t-1} (1 - b_i^*) \leq c \leq \sum_{i=1}^t (1 - b_i^*)$, then²

$$\varepsilon_i^* = \begin{cases} 1 - b_i^* & i \in [t-1], \\ c - \sum_{i=1}^{t-1} (1 - b_i^*) & i = t, \\ 0 & i \in [t+1 : H_v]; \end{cases} \tag{41}$$

(c) defining $\alpha_i^* = b_i^* + \varepsilon_i^*$; and (d) the fact that $\boldsymbol{\alpha}^* = \mathbf{b}^* + \boldsymbol{\varepsilon}^*$ is such that $\boldsymbol{\alpha}^* \in \mathcal{A}_\tau$; this is easily verifiable since:

- $b_i^* \geq 0$ for all $i \in [H_v]$ and $\varepsilon_i^* \geq 0$ for all $i \in [H_v]$ (by the construction in (41)) and hence, $\alpha_i^* = b_i^* + \varepsilon_i^* \geq 0$ for all $i \in [H_v]$.
- With the construction of ε_i^* in (41), for all $i \in [t-1]$, we have that $\alpha_i^* = b_i^* + \varepsilon_i^* = 1$ and for all $i \in [t+1 : H_v]$, we have that $\alpha_i^* = b_i^* + \varepsilon_i^* = b_i^* \leq 1$. Moreover, for $i = t$, we have that

$$\begin{aligned}
\alpha_t^* &= b_t^* + \varepsilon_t^* = b_t^* + c - \sum_{i=1}^{t-1} (1 - b_i^*) \\
&\leq b_t^* + \sum_{i=1}^t (1 - b_i^*) - \sum_{i=1}^{t-1} (1 - b_i^*) \\
&= b_t^* + (1 - b_t^*) + \sum_{i=1}^{t-1} (1 - b_i^*) - \sum_{i=1}^{t-1} (1 - b_i^*) = 1, \tag{42}
\end{aligned}$$

where the inequality follows since $c \leq \sum_{i=1}^t (1 - b_i^*)$. Thus, $\alpha_i^* = b_i^* + \varepsilon_i^* \leq 1$, for all $i \in [H_v]$.

- With the construction of ε_i^* in (41), it holds that

$$\sum_{i=1}^{H_v} \alpha_i^* = \sum_{i=1}^{H_v} (b_i^* + \varepsilon_i^*) = \sum_{i=1}^{H_v} b_i^* + \sum_{i=1}^{H_v} \varepsilon_i^* = \tau - c + \sum_{i=1}^{t-1} (1 - b_i^*) + c - \sum_{i=1}^{t-1} (1 - b_i^*) = \tau. \tag{43}$$

This concludes the proof of Lemma 4. ■

²Note that such a $t \in [H_v]$ always exists since $c = \tau - \sum_{i=1}^{H_v} b_i^* \leq H_v - \sum_{i=1}^{H_v} b_i^* = \sum_{i=1}^{H_v} (1 - b_i^*)$.

APPENDIX F

PROOF OF (d) IN (18)

Lemma 5. For a vector $\alpha \oslash \mathbf{h}$ of length H_v , let $\pi(\alpha \oslash \mathbf{h})$ be defined element-wise as follows,

$$\pi(i) := \left\{ j \in [H_v] : \frac{\alpha_j}{h_j} \text{ is the } i\text{-th maximum element in the vector } \alpha \oslash \mathbf{h} \right\}, \quad \forall i \in [H_v]. \quad (44)$$

Then, a solution to the following optimization problem

$$P_1 : \min_{\mathbf{K} \in \mathcal{K}} \sum_{i=1}^{H_v} \frac{\alpha_i}{h_i} (h_i - K_i) \quad (45)$$

is denoted as $\mathbf{K}^*$ and it has elements given by

$$K_{\pi(i)}^* = \min \left\{ K - \sum_{j=1}^{i-1} K_{\pi(j)}^*, h_{\pi(i)} \right\}, \quad i \in [H_v]. \quad (46)$$

Proof: We start by noting that the optimization problem P_1 in (45) can be rewritten as

$$P_1 : \sum_{i=1}^{H_v} \alpha_i - \max_{\mathbf{K} \in \mathcal{K}} \sum_{i=1}^{H_v} \frac{\alpha_i}{h_i} K_i. \quad (47)$$

Now, observe that by the definition of $\pi(i)$ in (44) all the coefficients $\frac{\alpha_i}{h_i}$, $i \in [H_v]$, which are non-negative rational numbers, are sorted in descending order, i.e., $\frac{\alpha_{\pi(1)}}{h_{\pi(1)}} \geq \frac{\alpha_{\pi(2)}}{h_{\pi(2)}} \geq \dots \geq \frac{\alpha_{\pi(H_v)}}{h_{\pi(H_v)}}$. This implies that an optimal solution for (47) would first fill $K_{\pi(1)}^*$ with its maximum value, i.e., $K_{\pi(1)}^* = \min \{ h_{\pi(1)}, K \}$. Then, it would fill $K_{\pi(2)}^*$ as $K_{\pi(2)}^* = \min \{ h_{\pi(2)}, K - K_{\pi(1)}^* \}$, and so on until $K_{\pi(H_v)}^*$. Thus, an optimal solution for (47) is given by (46). This concludes the proof of Lemma 5. $\blacksquare$

APPENDIX G

PROOF OF LEMMA 2

We let

$$\beta_i^* = \frac{M h_i}{H_e}, \quad \forall i \in [H_v], \quad (48a)$$

$$\lambda^* = 1 - \frac{K}{H_e}, \quad (48b)$$

$$\mu_i^* = \begin{cases} \left(1 - \frac{K}{H_e}\right) \sum_{j=1}^i h_{\pi(j)} & \text{if } i < t, \\ K \left(1 - \frac{\sum_{j=1}^i h_{\pi(j)}}{H_e}\right) & \text{if } i \geq t, \end{cases}, \quad \forall i \in [H_v - 1], \quad (48c)$$

$$\eta_i^* = \gamma_i^* = 0, \quad \forall i \in [H_v], \quad (48d)$$

where t is defined in Remark 1 and we show that the tuple $(\beta^*, \lambda^*, \mu^*, \eta^*, \gamma^*)$ above satisfies the KKT conditions associated with the optimization problem $P_{\text{inner}}(\pi)$ in (23). To see that the stationarity conditions in (26) hold with (48), we consider three separate cases and, for each of them, we use the expression of $K_{\pi(i)}^*$ given in (6), which equals K_j^π in (21) by definition, i.e., $K_j^\pi := K_{\pi(i)}^*$ where $\pi(i) = j$.

1) When $\pi(i) = j$ with $i < t$, the stationarity condition in (26) with (48) becomes

$$\frac{\partial}{\partial \beta_j} \mathcal{L}_\pi(\beta^*, \lambda^*, \mu^*, \eta^*, \gamma^*) = -\omega_j^\pi + \lambda^* + \frac{\mu_{\pi^{-1}(j)-1}^*}{h_j} - \frac{\mu_{\pi^{-1}(j)}^*}{h_j}$$

$$\begin{aligned}
&= -\omega_{\pi(i)}^\pi + \lambda^* + \frac{\mu_{i-1}^*}{h_{\pi(i)}} - \frac{\mu_i^*}{h_{\pi(i)}} \\
&\stackrel{(a)}{=} -\left(1 - \frac{h_{\pi(i)}}{h_{\pi(i)}}\right) + \left(1 - \frac{K}{H_e}\right) + \frac{\mu_{i-1}^*}{h_{\pi(i)}} - \frac{\mu_i^*}{h_{\pi(i)}} \\
&= \left(1 - \frac{K}{H_e}\right) + \frac{\mu_{i-1}^*}{h_{\pi(i)}} - \frac{\mu_i^*}{h_{\pi(i)}} \\
&\stackrel{(b)}{=} \left(1 - \frac{K}{H_e}\right) + \frac{1}{h_{\pi(i)}} \left(1 - \frac{K}{H_e}\right) (-h_{\pi(i)}) \\
&= 0,
\end{aligned} \tag{49}$$

where (a) follows by using (6) and λ^* in (48), and (b) uses μ^* in (48). Thus, for this case, the stationarity conditions are satisfied.

2) When $\pi(i) = j$ with $i = t$, the stationarity condition in (26) with (48) becomes

$$\begin{aligned}
\frac{\partial}{\partial \beta_j} \mathcal{L}_\pi(\beta^*, \lambda^*, \mu^*, \eta^*, \gamma^*) &= -\omega_j^\pi + \lambda^* + \frac{\mu_{\pi^{-1}(j)-1}^*}{h_j} - \frac{\mu_{\pi^{-1}(j)}^*}{h_j} \\
&= -\omega_{\pi(i)}^\pi + \lambda^* + \frac{\mu_{i-1}^*}{h_{\pi(i)}} - \frac{\mu_i^*}{h_{\pi(i)}} \\
&\stackrel{(a)}{=} -\left(1 - \frac{K - \sum_{j=1}^{i-1} h_{\pi(j)}}{h_{\pi(i)}}\right) + \left(1 - \frac{K}{H_e}\right) + \frac{\mu_{i-1}^*}{h_{\pi(i)}} - \frac{\mu_i^*}{h_{\pi(i)}} \\
&= \frac{K - \sum_{j=1}^{i-1} h_{\pi(j)}}{h_{\pi(i)}} - \frac{K}{H_e} + \frac{\mu_{i-1}^*}{h_{\pi(i)}} - \frac{\mu_i^*}{h_{\pi(i)}} \\
&\stackrel{(b)}{=} \frac{K - \sum_{j=1}^{i-1} h_{\pi(j)}}{h_{\pi(i)}} - \frac{K}{H_e} + \frac{1}{h_{\pi(i)}} \left(\left(1 - \frac{K}{H_e}\right) \sum_{j=1}^{i-1} h_{\pi(j)} - K \left(1 - \frac{\sum_{j=1}^i h_{\pi(j)}}{H_e}\right) \right) \\
&= 0,
\end{aligned} \tag{50}$$

where (a) follows by using (6) and λ^* in (48), and (b) uses μ^* in (48). Thus, for this case, the stationarity conditions are satisfied.

3) When $\pi(i) = j$ with $i > t$, the stationarity condition in (26) with (48) becomes

$$\begin{aligned}
\frac{\partial}{\partial \beta_j} \mathcal{L}_\pi(\beta^*, \lambda^*, \mu^*, \eta^*, \gamma^*) &= -\omega_j^\pi + \lambda^* + \frac{\mu_{\pi^{-1}(j)-1}^*}{h_j} - \frac{\mu_{\pi^{-1}(j)}^*}{h_j} \\
&= -\omega_{\pi(i)}^\pi + \lambda^* + \frac{\mu_{i-1}^*}{h_{\pi(i)}} - \frac{\mu_i^*}{h_{\pi(i)}} \\
&\stackrel{(a)}{=} -(1 - 0) + \left(1 - \frac{K}{H_e}\right) + \frac{\mu_{i-1}^*}{h_{\pi(i)}} - \frac{\mu_{\pi^{-1}(i)}^*}{h_{\pi(i)}} \\
&= -\frac{K}{H_e} + \frac{\mu_{i-1}^*}{h_{\pi(i)}} - \frac{\mu_i^*}{h_{\pi(i)}} \\
&\stackrel{(b)}{=} -\frac{K}{H_e} + \frac{1}{h_{\pi(i)}} \left(K \left(1 - \frac{\sum_{j=1}^{i-1} h_{\pi(j)}}{H_e}\right) - K \left(1 - \frac{\sum_{j=1}^i h_{\pi(j)}}{H_e}\right) \right) \\
&= 0,
\end{aligned} \tag{51}$$

where (a) follows by using (6) and λ^* in (48), and (b) uses μ^* in (48). Thus, for this case, the stationarity conditions are satisfied.

The above shows that the *stationarity* conditions are satisfied. For the *primal feasibility* conditions in (27), we have that $0 \leq \beta_i^* \leq 1$ for all $i \in [H_v]$. This is because $\beta_i^* = \frac{Mh_i}{H_e}$, which is non-negative, and $\beta_i^* = \frac{Mh_i}{H_e} \leq \frac{Mh_1}{H_e} \leq 1$, where the first inequality follows because of the assumption $h_1 \geq h_2 \geq \dots \geq h_{H_v}$ and the second inequality follows since we are in the regime $h_1 \leq H_e/M$. Moreover, since $\beta_i^* = \frac{Mh_i}{H_e}$, we also have that $\frac{\beta_i^*}{h_i} = \frac{M}{H_e}$ for all $i \in [H_v]$ and hence, $\beta^* \in \mathcal{B}_\pi$, i.e., the primal feasibility conditions are satisfied. It is an easy exercise to see that the *dual feasibility* conditions in (29) are also satisfied. Finally, the *complementary slackness* conditions in (30) are also trivially satisfied since $\frac{\beta_{\pi(i+1)}^*}{h_{\pi(i+1)}} - \frac{\beta_{\pi(i)}^*}{h_{\pi(i)}} = \frac{M}{H_e} - \frac{M}{H_e} = 0$ for all $i \in [H_v - 1]$, and $\eta_i^* = \gamma_i^* = 0$ for all $i \in [H_v]$.

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